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Descriptive Tables in R

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Descriptive Tables in R

- Summarize and organize data
- Give a simple overview of key data
- Help present results clearly

Why Are Descriptive Tables Important?

- Identify patterns and trends
- Detect inconsistencies or unusual values
- Compare different groups within the data
- Present results in a clear and accessible format

What Do Descriptive Tables Include?

- Categorical variables (e.g., team names)
  - Counts (n)
  - Percentages (%)
- Numerical variables (e.g., number of goals)
  - Mean
  - Median
  - Standard deviation (SD)

Janitor Package

- Used for cleaning and exploring data
- Helps create simple frequency tables
- Function used: tabyl()
- Can add percentages for clearer output

Example (Janitor)

```
liga %>%
  tabyl(Team) %>%
  adorn_percentages()
```

When to Use janitor?

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- Quick overview of categorical data
- Simple and readable tables
- Early stage of data exploration

Limitations:

- Mainly for simple summaries
- Does not provide advanced statistical measures
- Not intended for final publication tables

dplyr Package

- Core tool for data manipulation and transformation
- Allows flexible and detailed summaries
- Functions like group_by() , summarise() and count()

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What Does dplyr Do?

- Group data by categories
- Calculate summary statistics
- Create customized outputs
- Useful for numerical data within groups
- We can calculate multiple statistics at once

Example (dplyr)

```r
library(dplyr)

liga %>%
  group_by(Team) %>%
  summarise(
    mean_goals = mean(GF, na.rm = TRUE),
    sd_goals = sd(GF, na.rm = TRUE)
  )
```

When to Use dplyr?

- Need grouped summaries
- More control over calculations
- Preparing data for further analysis

Limitations:

- Output is not automatically formatted
- Tables are not presentation-ready
- Requires more manual coding than janitor

gtsummary Package

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- Creates publication-ready descriptive tables
- Automatically summarizes variables
- Produces clean and professional output
- Main function: tbl_summary()

What makes gtsummary different?

- Detects variable types automatically
- Applies appropriate summary statistics
- Formats tables for reporting

Example (gtsummary)

```
library(gtsummary)

liga %>%
  select(Team, GF ,GA) %>%
  tbl_summary(by = Team)
```

When to Use gtsummary?

- Preparing results for reports or presentations
- Clean and professional tables required
- Working with multiple variables

Limitations:

- Less flexible than dplyr for custom calculations
- Requires clean and prepared data
- Mainly used at the final stage

Classroom Question

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- liga dataset (bundesligR)

1. How often has each team appeared?
2. What is the average performance?
    - Points
    - Wins
    - Goals
3. How to present results clearly?

Solution

Step 1 – Data Preparation

```r
library(dplyr)
library(janitor)
library(gtsummary)

liga <- as_tibble(bundesligR)
```

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Solution

Step 2 – Frequency Table (janitor)

```
liga %>%
  tabyl(Team) %>%
  adorn_pct_formatting(digits = 1)
```

Solution

Step 3 – Summary Statistics (dplyr)

```
liga %>%
  group_by(Team) %>%
  summarise(
    seasons = n(),
    avg_points = mean(Points, na.rm = TRUE),
    avg_wins = mean(W, na.rm = TRUE),
    avg_goals = mean(GF, na.rm = TRUE)
  ) %>%
  arrange(desc(avg_points))
```

Solution

Step 4 – Final Table (gtsummary)

```r
liga %>%
  filter(Team %in% c(
    "FC Bayern Muenchen",
    "Borussia Dortmund",
    "Borussia Moenchengladbach"
  )) %>%
  select(Team, Points, W, D, L, GF, GA) %>%
  tbl_summary(
    by = Team,
    statistic = list(all_continuous() ~ "{mean} ({sd})"),
    digits = all_continuous() ~ 1
  )
```

Conclusion

- Descriptive tables are essential for understanding data
- janitor, dplyr, and gtsummary serve different roles
- janitor: quick overview of categorical data
- dplyr: flexible data analysis and summaries
- gtsummary: clean and publication-ready tables
- Combining these tools improves clarity and reporting

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Thank you for your attention